Dynamics: Exercises

Part I

Exercise 1
Three forces acting on an object are given by $\vec{F_1} = (2.00\vec{i} + 2.00\vec{j})$ N, $\vec{F_2} = (5.00\vec{i} - 3.00\vec{j})$ N, and $\vec{F_3} = (-45.0\vec{i})$ N. The object experiences an acceleration of magnitude $3.75m/s^2$.

1. What is the direction of the acceleration?

2. What is the mass of the object?

3. What are the velocity components of the object after $10.0s$?

Exercise 2
A force $\vec{F}$ applied to an object of mass m_1 produces an acceleration of $3.00m/s^2$. The same force applied to a second object of mass m_2 produces an acceleration of $1.00m/s^2$.
(a) What is the value of the ratio m_1/m_2?
(b) If m_1 and m_2 are combined, find their acceleration under the action of the force $\vec{F}$.

Exercise 3
A baseball player throws the ball with velocity $\vec{v} = v\vec{i}$ by uniformly accelerating it straight forward horizontally for a time interval $\Delta t = t - 0$. If the ball starts from rest, and suppose $\vec{v}$ and Δt given (a) through what distance d does it accelerate before its release? (b) What force F does the player exert on the ball? compute them as function of v and Δt.

Exercise 4
Two crates, one with mass 4.00 kg and the other with mass 6.00 kg, sit on the frictionless surface of a frozen pond, connected by a light rope (Figure 1). A woman wearing golf shoes (for traction) pulls horizontally on the 6.00 kg crate with a force $\vec{F}$ that gives the crate an acceleration of 2.90 m/s^2.

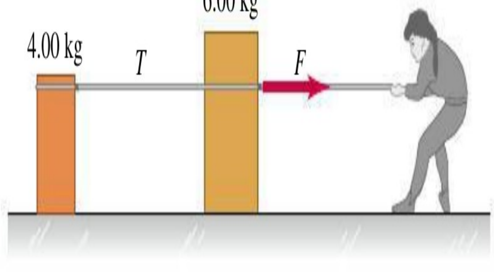
1. What is the acceleration of the 4.00 kg crate?

2. Draw a free-body diagram for the 4.00 kg crate. Use that diagram and Newton's second law to find the tension T in the rope that connects the two crates.

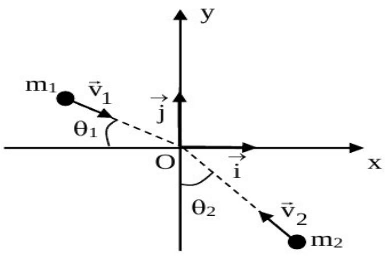
3. Draw a free-body diagram for the 6.00 kg crate. What is the direction of the net force on the 6.00-kg crate? Which is larger in magnitude, $\vec{T}$ or $\vec{F}$?

4. Use part (3) and Newton's second law to calculate the magnitude of $\vec{F}$.

Figure 1



Exercise 5
A prticle (1) of mass m_1 and a speed $\vec{v}_1$, is colliding another particle (2) of mass m_2 and a speed $\vec{v}_2$ (As show by Figure) and the system of the two particles is supposed isolated.
1-After collision the velocity of the particle (1) is $\vec{v\prime}_1 = (1.00\vec{i} + 2.00\vec{j})$ what is the velocity $\vec{v\prime}_2$ of the second particle.
2-If the two objects after collision are combined to one object find the components V_x and V_y of the combined object velocity $\vec{V}$. We give: $m_1 = 9\ kg$, $m_2 = 1\ kg$, $v_1 = 1\ m/s$, $v_2 = 3\ m/s$, $\theta_1 = \pi/6\ rad$,$\theta_2 = \pi/3\ rad$.



Exercise 6

Observed in the laboratory frame, a body of mass m_1 moving at speed v collides elastically with a stationary mass m_2. After the collision, the bodies move at angles θ_1 and θ_2 relative to the original direction of motion of m_1.

1. Find the velocity of the center of mass (CM) frame of m_1 and m_2.
2. Hence show that before the collision in the CM frame m_1 and m_2 are approaching each other, m_1 with speed $\frac{m_2 v}{m_1+m_2}$ and m_2 with speed $\frac{m_1 v}{m_1+m_2}$
3. In the CM frame after the collision m_1 moves off with speed $\frac{m_2 v}{m_1+m_2}$ at an angle θ to its original direction. Draw a diagram showing the direction and speed of m_2 in the CM frame after the collision.
4. Find an expression for the speed m_1 after the collision in the laboratory frame in terms of m_1, m_2, v and the angle θ.

Exercise 7

An open railway car of mass M is running on smooth horizontal rails under rain falling vertically down which it catches and retains in the car. If v_0 is the initial velocity of the car and k the mass of rain falling into the car per unit time, show that the distance traveled in time t is $(Mv_0/k)ln(1 + kt/M)$

Exercise 8

The point masses m and $2m$ lie along the x-axis, with m at the origin and $2m$ at $x = L$. A third point mass M is moved along the x-axis.

1. At what point is the net gravitational force on M due to the other two masses equal to zero?
2. Sketch the x-component of the net force on M due to m and $2m$, taking quantities to the right as positive. Include the regions $x < 0$, $0 < x < L$, and $x > L$. Be especially careful to show the behavior of the graph on either side of $x = 0$ and $x = L$.

Exercise 9

1. Let us consider that the moon motion around the earth is circular uniform of period T_1 and radius r_1. Find the moon acceleration.
2. A satellite of Jupiter has a circular motion of radius r_2 and a period T_2 relative to Jupiter.
a. Find the forces F_1 of the Earth on moon and F_2 of Jupiter on its satellite.
b. Calculate the ratio M_J/M_T of Jupiter mass on Earth mass.
Data: $r_1 = 3,84 \times 10^5\ km$, $T_1 = 27,3\ days$, $r_2 = 6,7 \times 10^5\ km$, $T_2 = 3.5\ days$.